SiTime

SITM | NASDAQ | 2026-10-07

Q4 2027 model ceiling: USD 246.17

Analytical Base 45/60 = 75 | Qualified technology bottleneck / Watchlist

20x forward-four-quarter GAAP operating profit including SBC, depreciation and acquisition amortization, less start-quarter note principal and remaining acquisition earnout, divided by end-window diluted common shares. No cash credit. Each ceiling includes its labeled quarter plus next three.

Coverage initiated at the owner's request. No holding was established by this research. Existing portfolio Base overrides remain protected.

All operating values are in USD m and shares in millions unless stated otherwise. Forecasts are analytical assumptions; issuer guidance and actual results are identified separately.

Writing-date snapshot: USD 711.06 per share (2026-10-06 close); 30.071 million common shares (2026-08-03 SEC 10-Q cover); estimated equity capitalization USD 21,382.3 million. Quote and disclosed share count have different dates; capitalization is their stated-date product, not a live market feed. This dated common-share snapshot is separate from forecast diluted shares.

Decision in the model

Active horizon	Earnings window	F4Q OP	Multiple	Diluted shares	Ceiling
Q4 2027	Q4 2027 through Q3 2028	468.5	20x	32.410	246.17

The ceiling is a model valuation, not a guaranteed trading price or a literal net-income P/E. Operating branches are weighted once. Where debt, preferred claims, conversion or equity floors make value nonlinear, the model values each coherent operating and funding state first, then weights those common-share outcomes.

Economic scope and matching claims

Consolidated common-owner GAAP operating profit; leases remain expensed; outstanding earnout and note principal deducted once. Equity floors are applied state by state before weighting. Debt repayment is the more conservative case than conversion in all modeled outcomes.. 20x forward-four-quarter GAAP operating profit including SBC, depreciation and acquisition amortization, less start-quarter note principal and remaining acquisition earnout, divided by end-window diluted common shares. No cash credit. Each ceiling includes its labeled quarter plus next three.

Financial claims and diluted shares follow the company-specific schedule below. Debt, preferred and minority claims are counted once. All financial model tables use USD m unless a different unit is stated.

Investment judgment

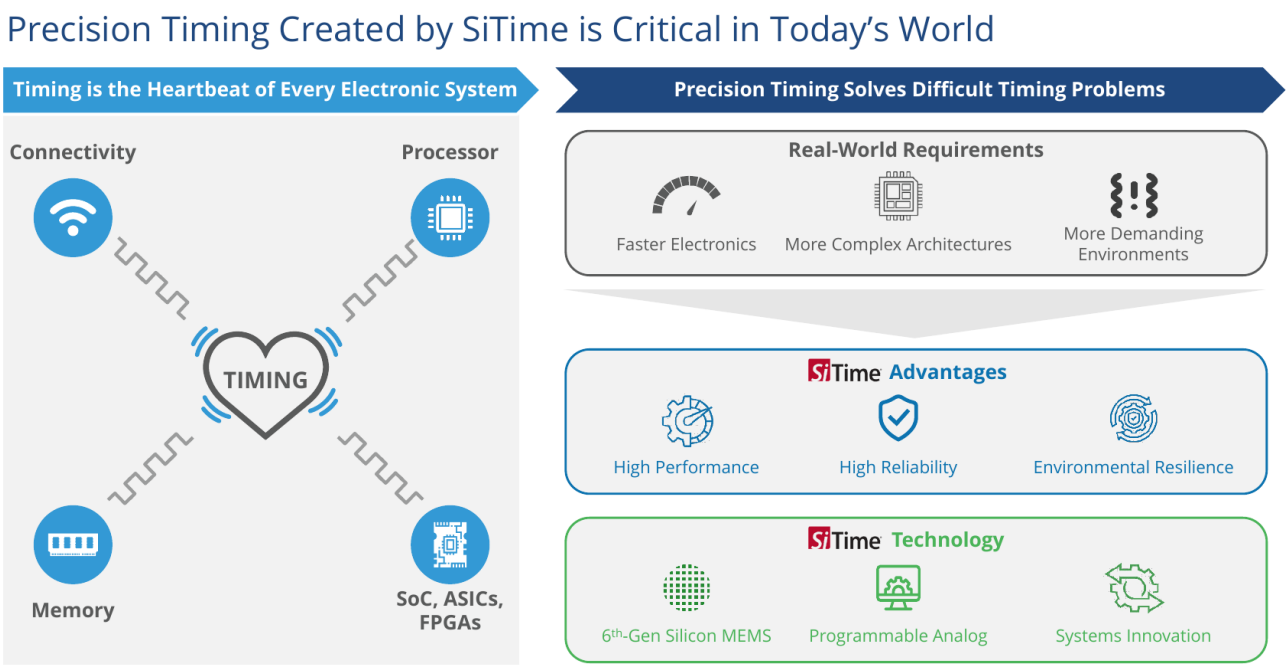
SiTime sells a small component whose failure can impair a much more expensive system. The opportunity is rising timing content, tighter performance requirements and a broader clock portfolio after the July acquisition of Renesas' Timing Products Division. The active Q4 2027 ceiling is **$246.17**, using 20x forward-four-quarter fully expensed GAAP operating profit, outstanding financial claims and end-window diluted shares. Analytical Base is **45/60, or 75%**. This is a qualified technology bottleneck with competitive and execution risk, rather than an exclusive supplier to all AI systems.

The distinction between product economics and shareholder earnings is unusually important here. An attractive incremental gross margin does not remove acquired-technology amortization, employee compensation, the purchase-financing claim or dilution. The model retains them. It also gives credible forward technical evidence a numerical contribution before every customer award becomes public. No analyst target or current stock-price multiple determines the ceiling.

Research date is October 7, 2026; Framework 8.0 applies. This is new coverage, not evidence of a holding. The model covers Q3 2026-Q4 2030; each of 15 ceiling quarters values itself and the following three quarters. Revenue decompositions beyond disclosed end-market totals are explicitly analytical estimates.

Business and engineering

An oscillator creates a reference frequency; clocks, buffers and synchronizers distribute or condition it. SiTime combines a silicon MEMS resonator with programmable analog circuitry, compensation and system design. The commercial moat depends on phase noise, jitter, frequency stability over temperature and aging, resilience, package size, programmability, reliability and qualification. A single favorable specification does not settle the comparison.

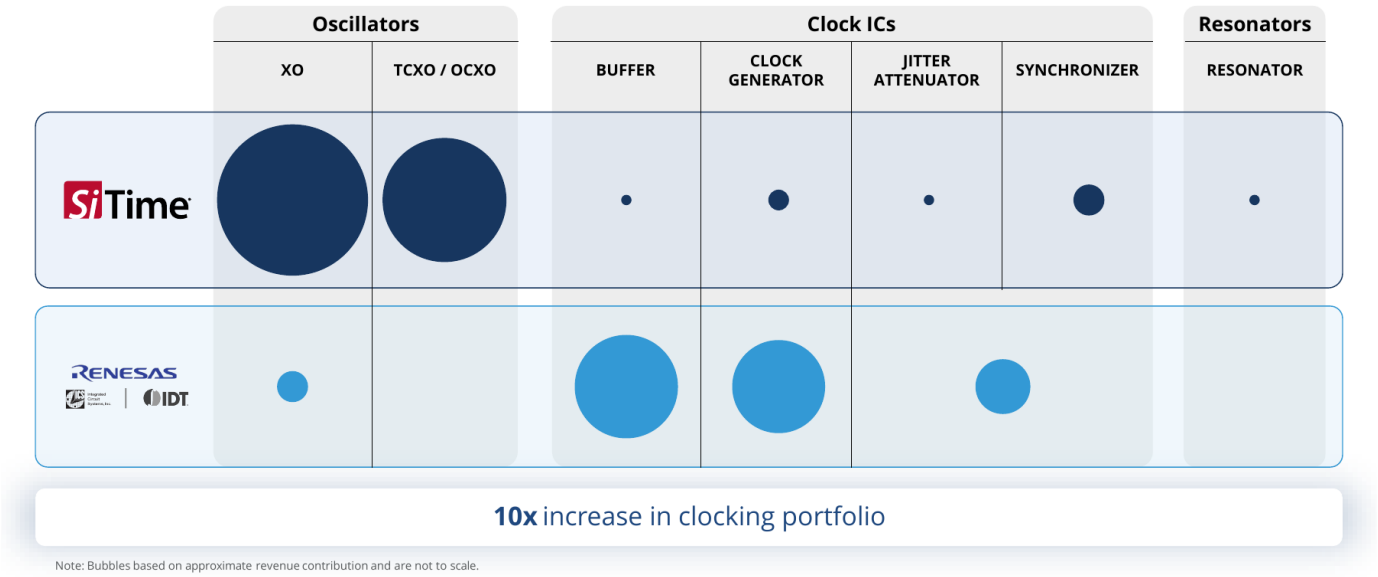


S1 | Issuer architecture: MEMS plus analog and systems design. Undated IR deck with 2025 reference data; conceptual map, not current guidance.

The product is valuable when its performance, delivery or reliability saves the customer more than the component premium. September management described prices ranging from roughly $0.35-0.40 to $50 outside military applications. This breadth makes average company ASP a poor proxy for advanced data-center performance. Customers can redesign around another qualified timing component, but that takes engineering effort and validation. The resulting barrier supports premium products; it does not eliminate competitive sourcing.

The acquired timing business

The July 1 transaction added a mature clock portfolio with approximately 550 products and 10,000 customers. It broadens SiTime beyond its stronger oscillator position into clock generation, buffering and synchronization. The February product map is useful for understanding complementarity. Its original financing and closing forecasts have been superseded by the completed transaction.



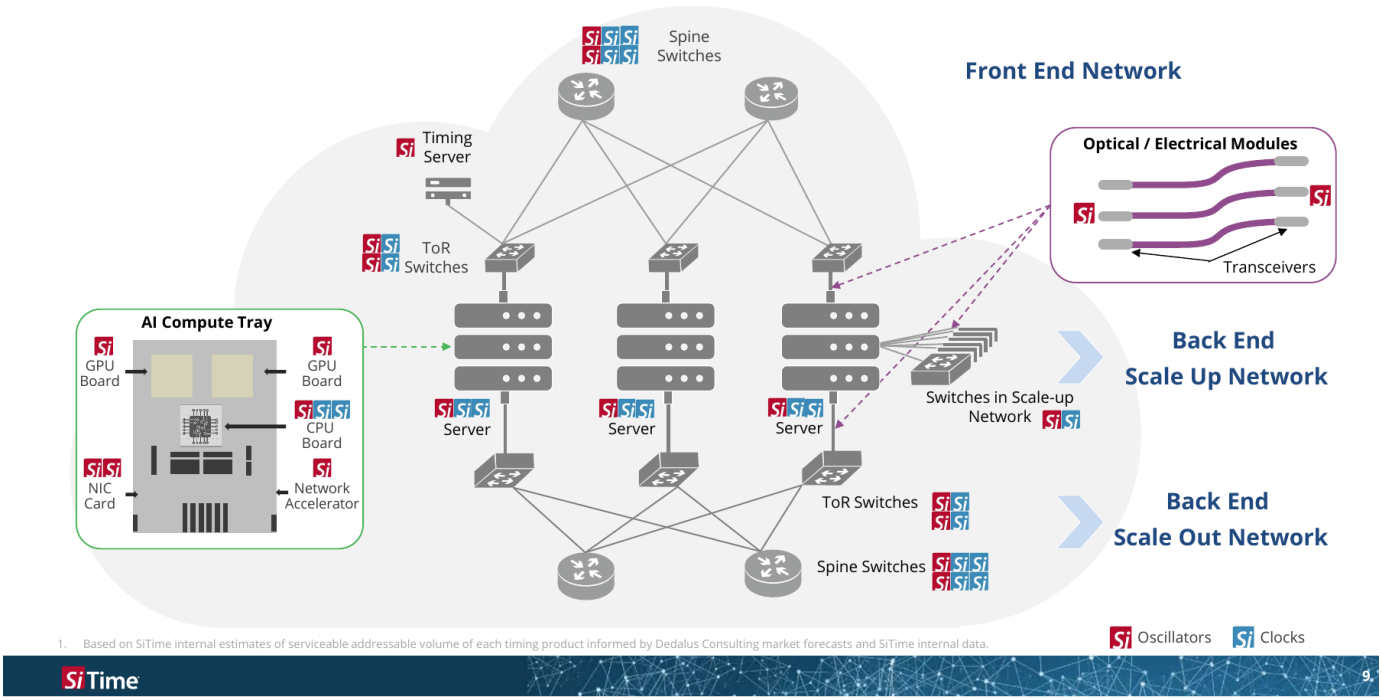
S2 | SiTime February 4 acquisition deck p7. Product complementarity; bubble sizes are expressly not to scale. Closing figures are updated in the text.

The acquired business produced $83.732 million of revenue in Q2 and $153.038 million in the first half, before consolidation into SiTime. Q3 guidance includes $85 million. In September, the CEO described potential 2027 growth above 50%, possibly 60-80%, and said supply through Renesas' transitional arrangements was constraining current shipments. This is a specific management forecast, not an order guarantee.

The reference assumes $440 million TPD revenue in 2027. A 70% probability branch adds $126 million conditional revenue, reaching $566 million if successful. The active increment is $88.2 million revenue and $48.51 million OP after 73% incremental GM and 18% incremental development, sales and support costs. The older $300 million post-close forecast is already overtaken by the current run rate and is never added again.

Demand reaches several parts of the rack

Timing sits beside compute chips, NICs, switches and optical/electrical modules. September management described $20-25 of timing content in a typical switch and several hundred dollars, sometimes above $1,000, in particular rack configurations. The word particular matters: rack architecture, card count and product mix vary widely.



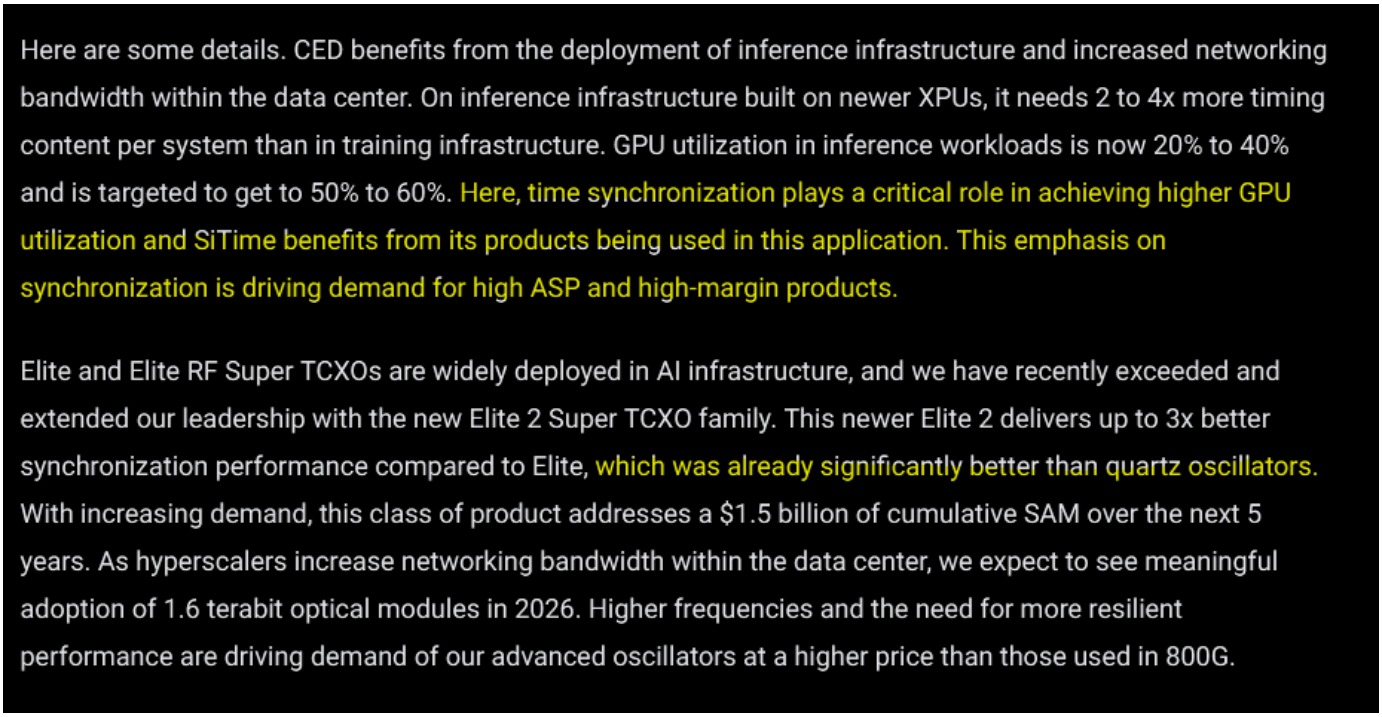
S3 | Issuer rack architecture p9. The old $1.2bn SAM label is historical; September management described a larger scope. No chart TAM is booked directly as revenue.

This older diagram's $1.2 billion SAM label is not a current company-revenue estimate. The September discussion used a wider roughly $3 billion CED opportunity. Neither becomes sales without unit deployment, qualification, realized content and supplier capture. The reference legacy CED vector rises from $620 million in 2027 to $1.3 billion in 2030. It already includes ordinary optical-speed transitions, existing design wins and more timing-intensive systems.

The same link may contain a laser from Lumentum, an optical module from AAOI or another supplier, a Marvell/Broadcom interface chip and a SiTime reference clock. Those are different component revenues. A change to CPO can move the timing location or replace a discrete component; it does not entitle every supplier to the whole networking bill.

Synchronization and useful compute

Management links inference to more timing content and better synchronization. Its May discussion described 2-4x timing content in certain inference systems and a goal of improving GPU utilization from 20-40% toward 50-60%. These are system-level statements. The report does not attribute the entire utilization improvement to one oscillator.



S4 | May 8 Irrational Analysis reproduction of SiTime Q1 call remarks. The $1.5bn opportunity is cumulative; utilization targets are not a measured product-only result.

Elite 2 was announced in May, with production expected in Q3. The company describes tighter thermal stability, compact packaging and improved synchronization. The quoted $1.5 billion opportunity is cumulative over several years, not an annual 2027 market. The active model instead uses a bounded 2027 proxy of 150,000 qualifying rack-equivalents, $300 incremental timing content and a 30% overlap deduction. At 65% probability, the net contribution is $20.475 million revenue and $10.647 million OP. Rack-equivalents normalize a mixed installed opportunity; they are not claimed physical shipment disclosures.

Testing the technical claim

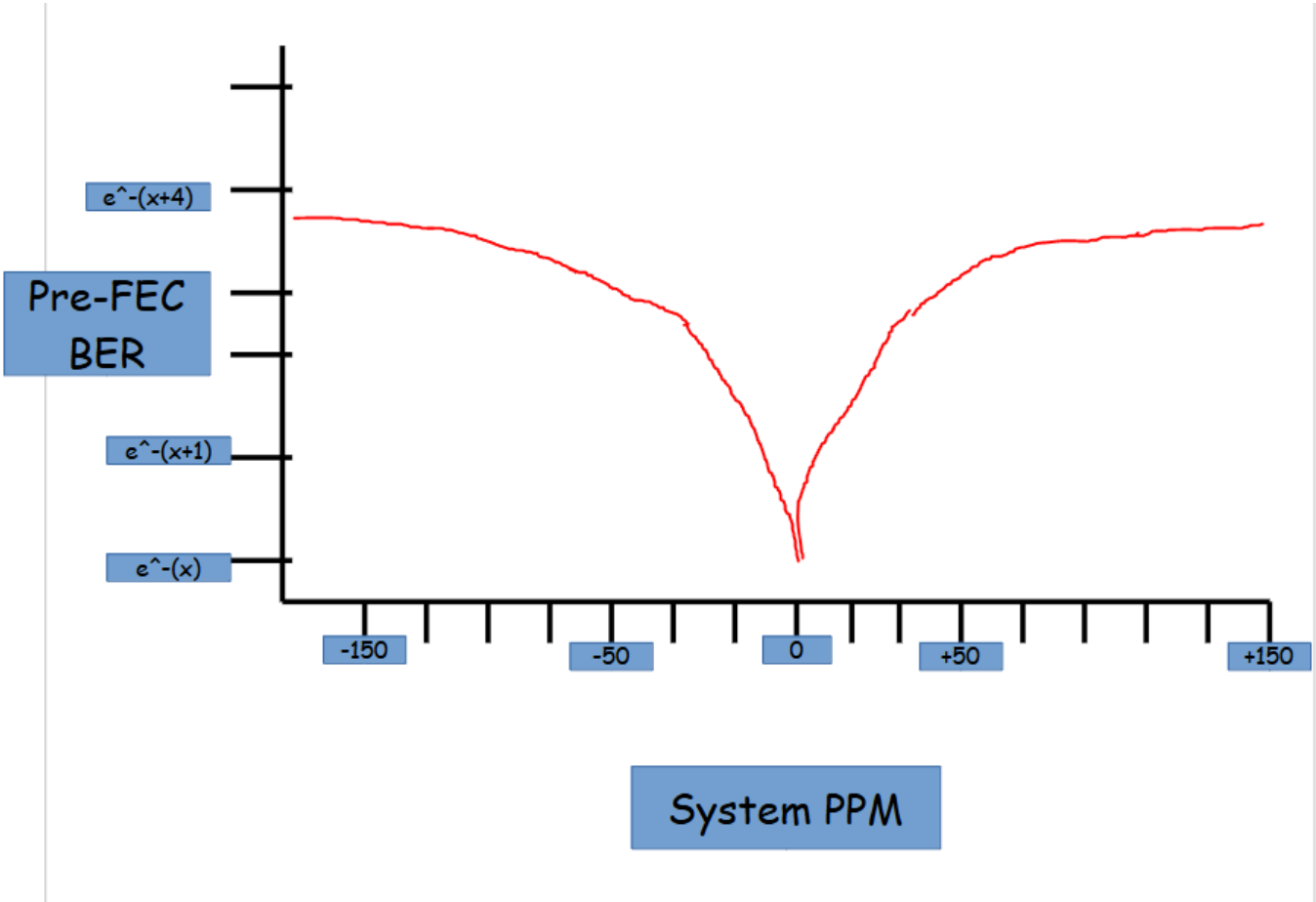
The supplied May 8 Irrational Analysis article makes a useful engineering argument about frequency offset, jitter and timing control, but its claim that quartz is finished is too strong. The reproduced transmitter table establishes a tolerance for one specification context. It does not establish a universal relationship between ±20 ppm, bit errors and customer revenue.

Table 162–11—Summary of transmitter specifications at TP2

Parameter	Subclause reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm[a]	GBd
Differential pk-pk voltage with Tx disabled (max)[b]	93.8.1.3	30	mV
DC common-mode voltage (max)[b]	93.8.1.3	1.9	V
AC common-mode peak-to-peak voltage (max) Low-frequency, VCM_{LF} Full-band, VCM_{FB}	162.9.4.4	 30 80	 mV mV
Differential pk-pk voltage, v_{di} (max)[b]	93.8.1.3	1200	mV
Effective return loss, ERL (min)	162.9.4.8	7.3	dB
Common-mode to common-mode return loss, *RLcc* (min)	162.9.4.9	See Equation (162–6)	dB
Common-mode to differential-mode return loss, *RLdc* (min)	162.9.4.10	See Equation (162–7)	dB
Transmitter steady-state voltage, v_f (min) Transmitter steady-state voltage, v_f (max)	162.9.4.1.2	0.387 0.6	V
Linear fit pulse peak ratio, R_{peak} (min)	162.9.4.1.2	0.397	—
Level separation mismatch ratio R_{LM} (min)	162.9.4.2	0.95	—
Transmitter output waveform absolute value of step size for all taps (min) absolute value of step size for all taps (max) value at minimum state for *c*(–3) (max) value at maximum state for *c*(–2) (min) value at minimum state for *c*(–1) (max) value at minimum state for *c*(0) (max) value at minimum state for *c*(1) (max)	 162.9.4.1.4 162.9.4.1.4 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5	 0.005 0.025 −0.06 0.12 −0.34 0.5 −0.2	 — — — — — — —

S5 | May 8 article reproduces a 53.125 GBd transmitter table with +/-50ppm limit. Exact source edition is unresolved; no universal +/-20ppm financial conversion is inferred.

Source-image quantity	Reading used	Model treatment
Lane symbol rate	53.125 GBd	Engineering context only
Displayed tolerance	±50 ppm	No universal ±20 ppm rule inferred
Edition and test implementation	Not identified in supplied excerpt	Remains a source gap



S6 | Irrational Analysis May 8 conceptual pre-FEC BER versus system PPM. Uncalibrated sketch; not measured test data or a revenue input.

The accompanying BER curve is a conceptual sketch without calibrated measured data or a specified receiver. It supports a hypothesis to test, not a quantified competitive benchmark. Clock recovery, transmitter/receiver frequency offset, PLL response and the reference-clock phase-noise spectrum all matter.

The current SiT9501 datasheet makes the measurement problem concrete. Under a legacy 12 kHz-20 MHz integration filter, its example compares 70 fs against 50 fs for another device. Under the stated 4-16A method, the example becomes 45 fs against 94 fs. These are not contradictory results: the filters weight noise differently. They also cannot be compared to an unrelated part's headline jitter without matching output frequency, voltage, signaling, temperature and measurement method.

"4-16A" Phase Jitter Methodology for SerDes Applications

Proper evaluation of reference clock (refclk) jitter is critical to optimize system performance in high-speed serial links.

Figure 5 shows how the traditional 12 kHz to 20 MHz analysis of filtering refclk jitter can mislead designers to select components that degrade rather than improve link performance. Using a more accurate filter analysis predicts a 50% reduction (45 vs 94 fs RMS) in jitter. Therefore, this datasheet replaces the legacy 12 kHz to 20 MHz filter analysis with a more accurate and established methodology adopted by several industry standards (e.g. PCI Express, CXL, UCIe) and implemented here as "4-16A" phase jitter. A brief overview follows.

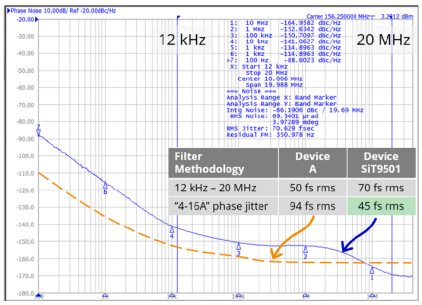


Figure 5. Two products analyzed with 2 different methodologies lead to opposite conclusions. The "4-16A" phase jitter methodology more accurately predicts in-system performance for modern SerDes applications.

Established in 1991 for SONET OC-48 line rates, the traditional 12 kHz to 20 MHz jitter filter served as a golden reference to evaluate refclk jitter for over 30 years. The filter is used is nearly all clock and timing datasheets today. However, the results it provides no longer correlate with system performance and can create suboptimum link performance. Sources of filter error include incorrect corner frequencies, unrealistic brick-wall roll offs and a lack of accounting for aliased phase noise. Errors of tens of femtoseconds are significant today and will become more significant as data rates increase. For these reasons, we recommend customers adopt the more accurate "4-16A" phase jitter methodology for SerDes applications.

The conventional refclk jitter analysis uses a band-pass filter, as shown in Figure 6, to extract the refclk contribution to jitter observed at the receiver. Historically the refclk jitter filter was arbitrarily applied to phase noise up to an offset equal to the refclk Nyquist frequency. However, this ignores higher-offset phase noise that aliases when the refclk is sampled by a PLL's digital phase detector. Studies have shown that extending the measured phase noise data flat to the third harmonic (or, twice the fundamental frequency in the offset-frequency axis) accurately estimates worst-case phase jitter[12]. Above the third harmonic, phase noise rolls off quickly and can be ignored.

Note:
12. "How to evaluate reference-clock phase noise in high-speed serial links", Signal Integrity Journal (May 3, 2019).

S7 | SiTime SiT9501 rev1.02, July 30 2026, p16. Same parts rank differently under different filters; 45fs and 70fs are different measurements.

Issuer illustrative comparison	Device A	SiT9501
Legacy 12 kHz-20 MHz filter	50 fs RMS	70 fs RMS
4-16A filter	94 fs RMS	45 fs RMS
Interpretation	Relative ranking changes with method	Not a universal competitive test

A 70 fs SiT9501 was announced in 2020; the datasheet revision history introduced 4-16A in 2023. The July 2026 revision lists symbol/disclaimer changes rather than a newly invented sub-100 fs part. Most decisively, the May earnings call acknowledges credible quartz suppliers at 1.6T and continued shared business. The model grants differentiated-product growth and a bounded repricing branch, but no additional monopoly margin.

Latest actuals and guidance

Q2 SiTime revenue was $157.432 million, gross profit $99.142 million and GAAP OP $8.215 million. R&D of $36.117 million, SG&A of $46.301 million and acquisition-related expenses of $8.509 million reconcile the operating result. Non-GAAP OP was $53.461 million. The $45.246 million difference largely comprises SBC, amortization and acquisition costs; those costs cannot be silently discarded when comparing valuation bases.

Q2 2026 disclosed measure	USD million
Revenue	157.432
GAAP gross profit	99.142
GAAP operating costs	90.927
GAAP operating profit	8.215
Non-GAAP operating profit	53.461
Total SBC	30.993
Operating cash flow, rounded management figure	40.0
Capital spending	12.9

Q3 combined revenue guidance is $285-295 million, including $85 million TPD and $200-210 million legacy SiTime. Non-GAAP GM is 68% ±1 point and non-GAAP opex $80-85 million. The model anchors to the midpoints, then expenses the acquisition amortization, SBC and a transition provision. Product-specific margins are estimated allocations calibrated to consolidated guidance, not reported segment margins.

The May and August calls show an expanding business and continuing investment. September updates the acquired-business growth outlook and flags supply constraints. Management's 25-30% long-term growth target is an across-cycle ambition, not a cap on the current cycle or a guarantee for every year.

Purchase accounting and full expenses

The transaction paid approximately $1,499.25 million cash plus 3.558691 million new shares. At the closing valuation, those shares represented $2,504.749 million; reported purchase consideration totaled approximately $4,006.106 million. Treating the $1.5 billion cash portion as the entire acquisition price would omit most consideration.

liabilities. The remaining approximately $24.2 million had not been recognized in SiTime's historical financial statements as of June 30, 2026 and is reflected as an adjustment to accumulated deficit.

(iii) Reflects the adjustment to record the acquired inventory at its estimated fair value, determined based on the estimated selling price of the inventory in the ordinary course of business, less costs to sell, resulting in a fair value equal to historical inventory cost plus an estimated gross profit margin calculated on a cost basis.

(iv) Represents the net adjustment to the estimated fair value of intangible assets acquired in the Acquisition. Preliminary identifiable intangible assets in the pro forma financial information are provided in the table below. The estimated fair values of developed technology and brand assets were determined using the relief-from-royalty method. The estimated fair values of customer contracts and related relationships and order backlog were determined using the multi-period excess earnings method. The amortization related to these identifiable intangible assets is reflected as a pro forma adjustment in the unaudited pro forma condensed combined income statement, as further described in Note 6(ii). The identifiable intangible assets are preliminary and subject to change upon finalization of the purchase price allocation.

The general categories of the acquired identifiable intangible assets are the following:

(In thousands)		**Fair value**	**Estimated Useful Life (in years)**
Developed technology	$	1,600,000	11
Customer contracts and related relationships		462,000	12
Order backlog		51,000	1
Brand assets		7,000	6
Software		26	1
Total identifiable intangible assets	**$**	**2,120,026**	
Historical intangible assets carrying value		35,599	
Pro forma adjustment	**$**	**2,084,427**	

(v) Reflects the elimination of Timing Product Business' historical right-of-use asset and lease liabilities of approximately $1.2 million upon remeasurement of the lease balances in accordance with ASC 842.

(vi) Reflects the recognition of a right-of-use asset of approximately $1.2 million and corresponding current and non-current lease liabilities upon remeasurement of the lease balances using SiTime's incremental borrowing rate in accordance with ASC 842.

(vii) Represents the adjustment to goodwill based on the purchase price allocation.

(In thousands)		**Amounts**
Goodwill resulting from the Acquisition	$	1,877,687
Less: Timing Product Business' historical goodwill		(382,299)
Pro forma adjustment	**$**	**1,495,388**

(viii) Reflects the elimination of Timing Product Business' historical equity as of the Closing Date.

Note 6. Transaction Accounting Adjustments to Unaudited Pro Forma Condensed Combined Income Statement

(i) Reflects the elimination of intercompany royalty revenue and expense arising from the licensing arrangement between Aura Semiconductor Pvt. Ltd. ("Aura") and Timing Product Business following the settlement of the related contract-based royalty intangible asset recorded as a balance sheet adjustment in Note 5(i) above.

(ii) Represents the adjustment to record elimination of historical amortization expense and recognition of new amortization expense related to acquired identifiable intangible assets based on the estimated fair value and the associated estimated useful life. Amortization expense is calculated based on the estimated fair value of each of the identifiable intangible assets and the associated estimated useful life as discussed in Note 5(iv). The amortization is based on the periods over which the economic benefits of the intangible assets are expected to be realized, which are subject to adjustment as additional information becomes available.

Amortization Expense ***(In thousands)***		**For the Six Months Ended June 30, 2026**		**For the Year Ended December 31, 2025**
Developed technology	$	70,000	$	140,000
Customer contracts and related relationships		19,261		38,522
Order backlog		-		51,000
Brand assets		583		1,167
Total identifiable intangible assets	**$**	**89,844**	**$**	**230,689**

S8 | SiTime September 9 8-K/A p31. Figures in USD thousands. Preliminary intangibles and pro-forma amortization schedule; $51m backlog expires over one year.

Preliminary acquired intangible	USD million	Stated life
Developed technology	1,600.000	11 years
Customer relationships	462.000	12 years
Order backlog	51.000	1 year
Brand	7.000	6 years
Software	0.026	1 year

The published pro-forma schedule implies $179.689 million recurring annual amortization before backlog, or $44.922 million quarterly. The model follows that schedule and separately expenses $12.75 million quarterly backlog amortization for the first four post-close quarters. It also retains legacy acquired amortization and a $1.37 million inventory step-up over the first year. The developed-technology fair value divided by its stated life does not equal the published annual expense; final purchase accounting is therefore an explicit sensitivity.

The Q3 model's pre-adjustment product GP matches the 68% non-GAAP midpoint. Purchase amortization, inventory step-up and COGS SBC then reduce GAAP GP. Operating expenses separately include non-GAAP guided spending, operating SBC and transition costs. Ordinary depreciation and rent remain within the operating

cost base. The model never subtracts a historical TPD carve-out amortization charge and then adds the entire new charge twice.

Shares, debt and cash obligations

August 1 common shares were 30.071036 million. June outstanding unvested RSU, PRSU and multi-year awards totaled approximately 1.297094 million. The model starts with 31.45 million fully diluted shares, including a small replacement/new-award reserve, and adds 0.12 million quarterly for future net employee dilution. It does not assume future buybacks. Q3's 32.8 million non-GAAP diluted guide is not explicitly decomposed; the difference is retained as a reconciliation gap rather than being claimed as proven convert dilution.

The $1.35 billion zero-coupon note matures June 15, 2031. Its initial conversion ratio represents 1.297485 million shares at roughly $1,040.47. The model checks both debt repayment and conversion in all 32 operating outcomes and all 15 valuation windows. Debt repayment gives the lower common-share value throughout the modeled path, so principal is deducted and conversion shares are excluded. A revised scenario crossing the threshold must repeat the check. The prepaid capped call is not a second cash asset.

The remaining $81.353 million Aura earnout is a separate claim, based on revenue through December 2028. Its exact payment schedule is not disclosed; the model pays it evenly over ten quarters and deducts the remaining claim at the start of each valuation window. Higher earned payments are a sensitivity. Operating leases stay expensed, so no second lease-capitalization deduction is made. An uncommenced California lease with $83.2 million undiscounted payments is covered by a rent reserve within growing opex, subject to final commencement details.

The post-transaction pro-forma cash starting point is $389.264 million, not June's $1.921 billion pre-payment cash balance. Across all 32 operating paths, minimum subsequent quarter-end cash is about $424.174 million under the stated assumptions. The bridge includes 6% revenue capex, ordinary depreciation and amortization, SBC, a 20% incremental working-capital requirement, 18% tax on positive OP, employee tax withholding beginning at $45 million quarterly and the earnout payments. It excludes new acquisitions, buybacks and revolver proceeds.

SBC is retained in operating profit. Its noncash add-back in the funding bridge is accompanied by employee withholding cash outflows and a diluted share schedule. The $200 million undrawn revolver is potential contingent liquidity, not cash already owned. Debt maturity is outside the modeled horizon; strong modeled cash accumulation does not authorize ignoring the obligation.

Supply, pricing and capacity response

Bosch fabricates SiTime MEMS structures under a long relationship; management expects renewal of the agreement reaching its 2027 date. An expected renewal is not a signed extension. Mature analog process nodes at TSMC and outsourced assembly/test also matter. TPD manufacturing and testing remain partly dependent on Renesas transitional services.

These facts create two opposing effects. Qualified supply and customer delay costs support premium pricing, while foundry/test expansion and credible quartz alternatives can relieve scarcity. The model assigns 45% probability to a 2% premium on 60% of eligible legacy CED sales, with 50% removed for mix already modeled. Incremental expediting/support costs consume part of that premium. It adds only $1.256 million 2027 OP; the much larger growth contribution comes from actual product demand and content.

The reference is not nameplate production capacity. No complete public good-unit capacity or yield series was found. Revenue forecasts remain constrained by observed shipments, management's next-quarter guidance and a staged ramp; the model does not infer saleable output from factory floor area.

New sockets and non-AI demand

Embedded timing and substrate/chiplet integration can bring the reference source closer to the consuming device. September management suggested a $1-1.5 billion incremental SAM and earliest meaningful activity in 2028-29. The model assigns 35% probability to a bounded new-socket branch, removes 40% for substitution/reference overlap and retains an $8 million annual launch expense. Early 2028 weighted OP is slightly negative even though revenue contributes positively.

Legacy automotive, industrial, aerospace and defense revenue grows to $150 million in 2027 and $275 million in 2030. Mobile/IoT/consumer grows to $220 million and then $295 million, with first-quarter seasonality. These vectors already absorb modem adoption and a measured defense/industrial expansion. Lifetime design-win funnels, satellite counts and distant humanoid market forecasts are not booked as separate annual revenue.

Risks and paired watches

Four scored risk classes apply. Competition and customer-cycle risk include price pressure, alternative quartz solutions, customer concentration and AI-capex digestion. Supplier dependence includes Bosch renewal, TSMC and outsourced test, plus Renesas transitional supply. Acquisition execution includes integration, preliminary amortization, the true standalone cost base and product-roadmap investment. Capital structure and dilution include the convertible claim, earnout, employee awards and withholding cash.

The paired watches are SiTime with AAOI/LITE/COHR for optical-speed transitions, with MRVL/AVGO/NVIDIA for switch and compute architecture, and with Keysight for system-level validation. A timing win is a content opportunity in those systems, not proof that a particular optical supplier loses. Rising synchronization value also does not prove every inference workload requires the same expensive timing solution.

Microchip and Japanese quartz specialists remain competitive research references; supplier discovery must distinguish already-covered companies from holdings before a separate initiation. This report creates no additional stock rows beyond SiTime.

Valuation, sensitivities and next evidence

Active 2027 revenue is $1.509 billion and GAAP OP $257.298 million. The model reaches $3.025 billion revenue and $1.117 billion OP in 2030. The later margin requires scale, high-value product mix and controlled operating investment; it is not current guidance. The complete vector, cost, probability and share tables follow.

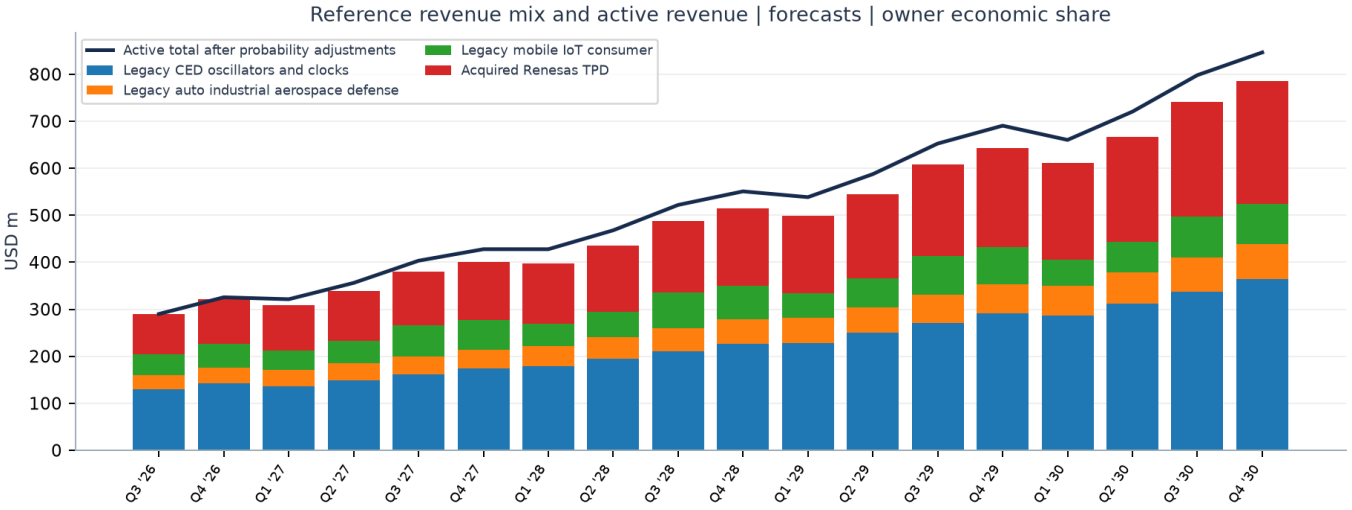
The Q4 2027 ceiling values Q4 2027 through Q3 2028. The acquisition's first-year backlog amortization rolls off before this window, but recurring purchased-technology amortization remains. The 20x convention is the default for new operating-profit coverage. Equity floors are applied to each coherent state before weighting, which matters in the earliest windows.

Near-term tests are the first combined results, TPD shipments versus the $85 million Q3 anchor, the full GAAP acquisition accounting and the diluted-share reconciliation. Through 2027, watch supply continuity, receivables and inventory, Elite2 production, product/customer mix and Bosch renewal. The probability ledger gives precommitted increases and reductions. A new technical claim should change the model only through its incremental qualified volume, price, cost, timing or risk channel.

Score arithmetic

Category	Points	Evidence / treatment
Core thesis	18	Differentiated qualified precision timing with credible quartz alternatives.
Cycle	8	Actual AI and acquired clock shipments are accelerating; concentration can cause digestion.
TAM	8	More endpoints, performance and timing content; incremental socket opportunities require qualification.
Catalysts	4	TPD supply expansion, Elite2 production, modem cycle and2027 supplier renewal.
Macro	9	AI connectivity, aerospace and industrial demand with trade and export constraints.
Alpha	2	Engineering detail matters but high-profile platform already followed.
Risk classes	-4	Four risk classes: competition/customer cycle; supplier dependence; acquisition/accounting execution; capital structure and dilution.
Total	45/60	Rounded Base 75. Both artifact and new workbook row use this score; this is not a manual override.

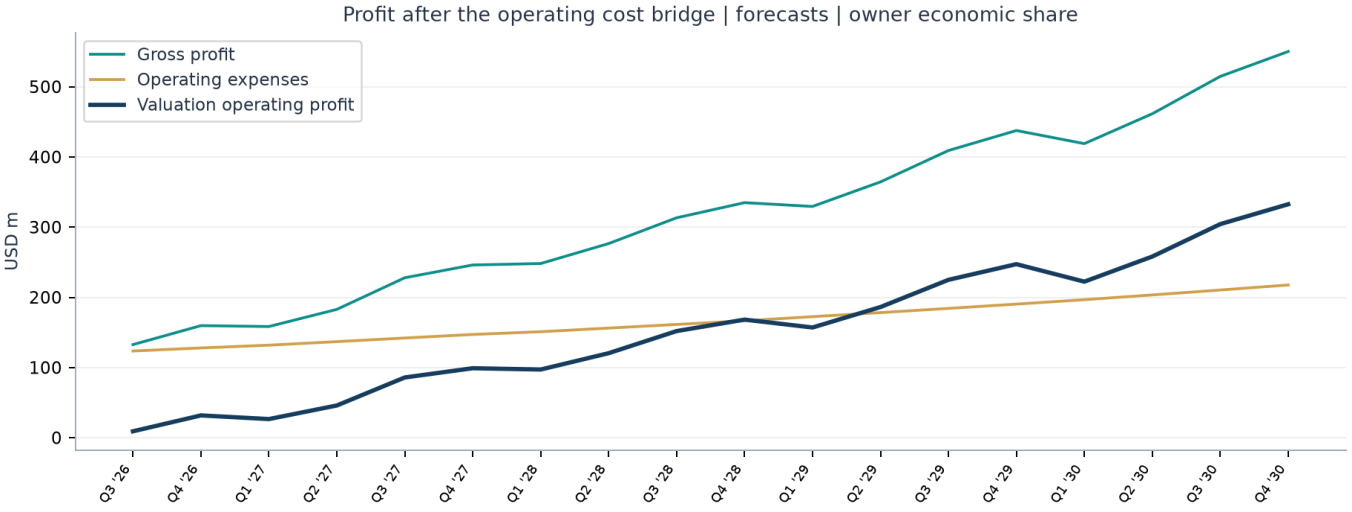
Revenue and business mix



Reference business vectors sum to the reference; the active total adds only probability-weighted net changes. Product allocations may be analytical proxies where issuers do not report product margins.

Quarter	Reference revenue	Weighted net change	Active revenue
Q3 2026	290.0	0.0	290.0
Q4 2026	322.0	3.5	325.5
Q1 2027	309.5	11.9	321.4
Q2 2027	338.8	17.3	356.1
Q3 2027	380.6	22.7	403.3
Q4 2027	401.1	26.7	427.8
Q1 2028	398.1	29.7	427.8
Q2 2028	435.4	32.6	468.0
Q3 2028	487.1	34.9	522.0
Q4 2028	514.4	36.4	550.8
Q1 2029	499.0	39.6	538.5
Q2 2029	545.3	42.2	587.5
Q3 2029	607.7	44.9	652.6
Q4 2029	643.1	47.5	690.6
Q1 2030	611.0	49.3	660.3
Q2 2030	667.3	53.1	720.4
Q3 2030	741.1	57.0	798.1
Q4 2030	785.6	60.8	846.4

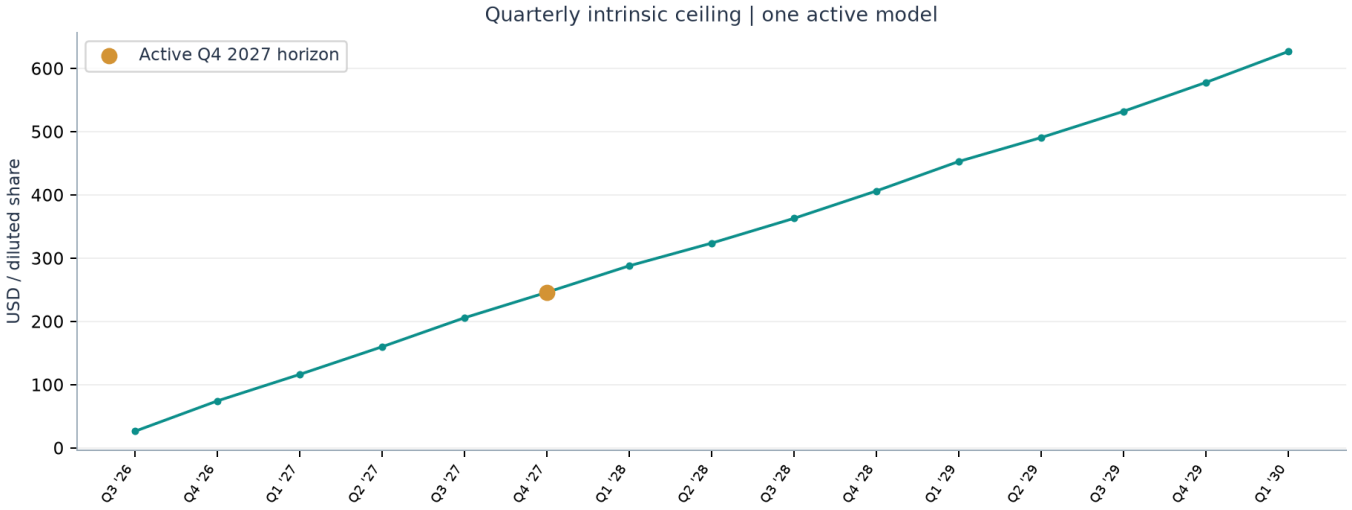
Gross profit, operating expenses and earnings



Operating profit equals gross profit less operating expenses and any separately identified attribution reserve. Figures are forecasts, not reported historical quarterly outcomes.

Quarter	Revenue	GP	Opex	Valuation OP	OP margin
Q3 2026	290.0	132.6	123.5	9.1	3.2%
Q4 2026	325.5	159.7	127.9	31.8	9.8%
Q1 2027	321.4	158.4	131.8	26.6	8.3%
Q2 2027	356.1	182.8	136.9	45.9	12.9%
Q3 2027	403.3	227.9	142.0	85.8	21.3%
Q4 2027	427.8	246.1	147.1	99.0	23.1%
Q1 2028	427.8	248.1	151.0	97.1	22.7%
Q2 2028	468.0	276.6	156.2	120.4	25.7%
Q3 2028	522.0	313.3	161.4	151.9	29.1%
Q4 2028	550.8	334.9	166.6	168.2	30.5%
Q1 2029	538.5	329.5	172.4	157.0	29.2%
Q2 2029	587.5	364.4	178.2	186.1	31.7%
Q3 2029	652.6	409.1	184.2	224.9	34.5%
Q4 2029	690.6	437.6	190.4	247.2	35.8%
Q1 2030	660.3	418.9	196.6	222.3	33.7%
Q2 2030	720.4	461.5	203.4	258.1	35.8%
Q3 2030	798.1	514.7	210.4	304.3	38.1%
Q4 2030	846.4	550.2	217.6	332.6	39.3%

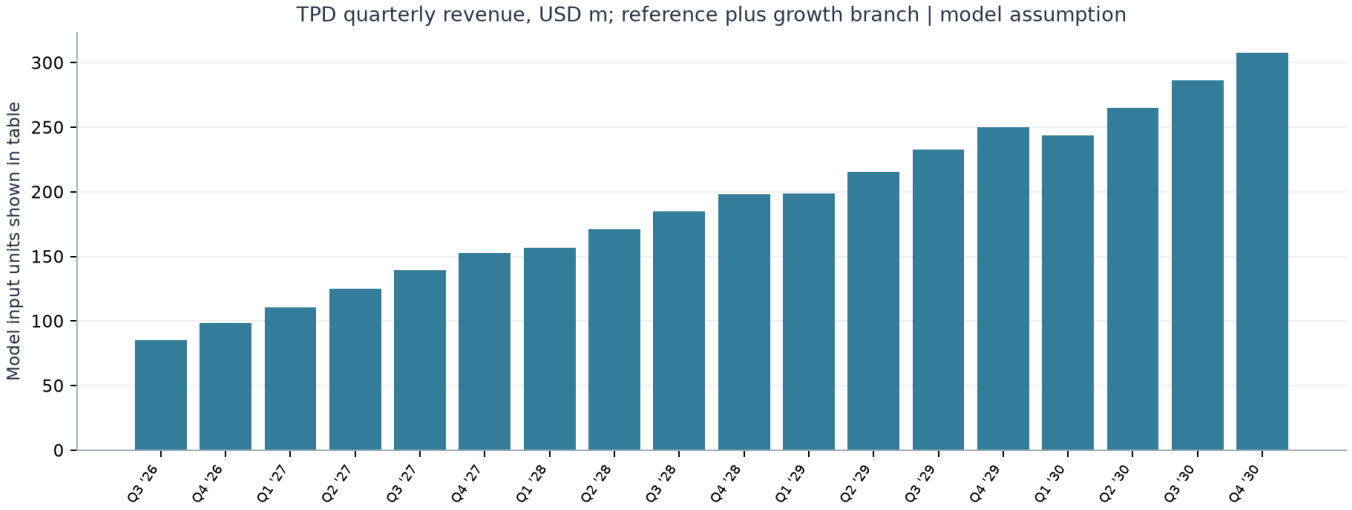
The complete ceiling path



Quarter Q prices Q plus the next three calendar quarters. All fifteen ceilings link to the same operating rows used in the workbook. Outer quarters have lower evidence visibility.

Quarter	Four-quarter earnings window	F4Q OP	Multiple	Shares	Ceiling
Q3 2026	Q3 2026 to Q2 2027	113.4	20x	31.810	26.50
Q4 2026	Q4 2026 to Q3 2027	190.1	20x	31.930	74.49
Q1 2027	Q1 2027 to Q4 2027	257.3	20x	32.050	116.41
Q2 2027	Q2 2027 to Q1 2028	327.9	20x	32.170	160.11
Q3 2027	Q3 2027 to Q2 2028	402.4	20x	32.290	205.91
Q4 2027	Q4 2027 to Q3 2028	468.5	20x	32.410	246.17
Q1 2028	Q1 2028 to Q4 2028	537.7	20x	32.530	288.09
Q2 2028	Q2 2028 to Q1 2029	597.6	20x	32.650	323.96
Q3 2028	Q3 2028 to Q2 2029	663.3	20x	32.770	363.14
Q4 2028	Q4 2028 to Q3 2029	736.3	20x	32.890	406.45
Q1 2029	Q1 2029 to Q4 2029	815.3	20x	33.010	453.08
Q2 2029	Q2 2029 to Q1 2030	880.6	20x	33.130	490.86
Q3 2029	Q3 2029 to Q2 2030	952.6	20x	33.250	532.36
Q4 2029	Q4 2029 to Q3 2030	1,031.9	20x	33.370	578.03
Q1 2030	Q1 2030 to Q4 2030	1,117.3	20x	33.490	626.92

The operating constraint behind the forecast



The driver series is a stated model assumption, not a disclosed commitment. It must be evaluated with the qualification, customer capture, cost and overlap discussion.

Quarter	TPD quarterly revenue, USD m; reference plus growth branch	Status
Q3 2026	85.000	Guidance anchored model
Q4 2026	98.500	Model
Q1 2027	110.800	Model
Q2 2027	125.200	Model
Q3 2027	139.600	Model
Q4 2027	152.600	Model
Q1 2028	156.700	Model
Q2 2028	171.200	Model
Q3 2028	185.000	Model
Q4 2028	198.100	Model
Q1 2029	198.600	Model
Q2 2029	215.700	Model
Q3 2029	232.800	Model
Q4 2029	249.900	Model
Q1 2030	243.500	Model
Q2 2030	265.000	Model
Q3 2030	286.500	Model
Q4 2030	308.000	Model

Quarterly revenue-vector derivation

All rows below are reference inputs. Gross profit = vector revenue x the stated gross margin. The next table applies the separately weighted evidence deltas. A product-level proxy is a model assumption even when the group total is calibrated to guidance.

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q3 2026	Legacy CED oscillators and clocks	130.0	65.1%	84.6
Q3 2026	Legacy auto industrial aerospace defense	30.0	61.0%	18.3
Q3 2026	Legacy mobile IoT consumer	45.0	59.0%	26.5
Q3 2026	Acquired Renesas TPD	85.0	3.7%	3.2
Q4 2026	Legacy CED oscillators and clocks	142.0	66.5%	94.5
Q4 2026	Legacy auto industrial aerospace defense	34.0	62.1%	21.1
Q4 2026	Legacy mobile IoT consumer	51.0	60.0%	30.6
Q4 2026	Acquired Renesas TPD	95.0	11.5%	10.9
Q1 2027	Legacy CED oscillators and clocks	136.4	66.5%	90.6
Q1 2027	Legacy auto industrial aerospace defense	34.5	62.3%	21.5
Q1 2027	Legacy mobile IoT consumer	41.8	60.0%	25.1
Q1 2027	Acquired Renesas TPD	96.8	12.7%	12.3
Q2 2027	Legacy CED oscillators and clocks	148.8	67.0%	99.7
Q2 2027	Legacy auto industrial aerospace defense	36.0	62.4%	22.5
Q2 2027	Legacy mobile IoT consumer	48.4	60.0%	29.0
Q2 2027	Acquired Renesas TPD	105.6	17.8%	18.8
Q3 2027	Legacy CED oscillators and clocks	161.2	67.4%	108.7
Q3 2027	Legacy auto industrial aerospace defense	39.0	62.6%	24.4
Q3 2027	Legacy mobile IoT consumer	66.0	60.0%	39.6
Q3 2027	Acquired Renesas TPD	114.4	33.5%	38.3
Q4 2027	Legacy CED oscillators and clocks	173.6	67.9%	117.8
Q4 2027	Legacy auto industrial aerospace defense	40.5	62.7%	25.4
Q4 2027	Legacy mobile IoT consumer	63.8	60.0%	38.3
Q4 2027	Acquired Renesas TPD	123.2	36.4%	44.8
Q1 2028	Legacy CED oscillators and clocks	178.2	68.1%	121.3
Q1 2028	Legacy auto industrial aerospace defense	43.7	62.8%	27.5
Q1 2028	Legacy mobile IoT consumer	47.5	60.0%	28.5
Q1 2028	Acquired Renesas TPD	128.7	38.0%	48.9

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q2 2028	Legacy CED oscillators and clocks	194.4	68.5%	133.2
Q2 2028	Legacy auto industrial aerospace defense	45.6	63.0%	28.7
Q2 2028	Legacy mobile IoT consumer	55.0	60.0%	33.0
Q2 2028	Acquired Renesas TPD	140.4	41.0%	57.6
Q3 2028	Legacy CED oscillators and clocks	210.6	68.9%	145.2
Q3 2028	Legacy auto industrial aerospace defense	49.4	63.1%	31.2
Q3 2028	Legacy mobile IoT consumer	75.0	60.0%	45.0
Q3 2028	Acquired Renesas TPD	152.1	43.5%	66.2
Q4 2028	Legacy CED oscillators and clocks	226.8	69.3%	157.1
Q4 2028	Legacy auto industrial aerospace defense	51.3	63.3%	32.5
Q4 2028	Legacy mobile IoT consumer	72.5	60.0%	43.5
Q4 2028	Acquired Renesas TPD	163.8	45.7%	74.9
Q1 2029	Legacy CED oscillators and clocks	228.8	69.8%	159.6
Q1 2029	Legacy auto industrial aerospace defense	52.9	63.4%	33.5
Q1 2029	Legacy mobile IoT consumer	52.2	60.0%	31.3
Q1 2029	Acquired Renesas TPD	165.0	46.0%	75.9
Q2 2029	Legacy CED oscillators and clocks	249.6	70.1%	175.0
Q2 2029	Legacy auto industrial aerospace defense	55.2	63.5%	35.1
Q2 2029	Legacy mobile IoT consumer	60.5	60.0%	36.3
Q2 2029	Acquired Renesas TPD	180.0	48.3%	87.0
Q3 2029	Legacy CED oscillators and clocks	270.4	70.4%	190.5
Q3 2029	Legacy auto industrial aerospace defense	59.8	63.7%	38.1
Q3 2029	Legacy mobile IoT consumer	82.5	60.0%	49.5
Q3 2029	Acquired Renesas TPD	195.0	50.3%	98.1
Q4 2029	Legacy CED oscillators and clocks	291.2	70.7%	205.9
Q4 2029	Legacy auto industrial aerospace defense	62.1	63.8%	39.6
Q4 2029	Legacy mobile IoT consumer	79.8	60.0%	47.9
Q4 2029	Acquired Renesas TPD	210.0	52.0%	109.2
Q1 2030	Legacy CED oscillators and clocks	286.0	70.8%	202.5
Q1 2030	Legacy auto industrial aerospace defense	63.2	64.0%	40.5
Q1 2030	Legacy mobile IoT consumer	56.0	60.0%	33.6

Quarter	Revenue vector	Revenue	Gross margin	Gross profit
Q1 2030	Acquired Renesas TPD	205.7	51.6%	106.2
Q2 2030	Legacy CED oscillators and clocks	312.0	71.0%	221.5
Q2 2030	Legacy auto industrial aerospace defense	66.0	64.0%	42.2
Q2 2030	Legacy mobile IoT consumer	64.9	60.0%	38.9
Q2 2030	Acquired Renesas TPD	224.4	53.5%	120.0
Q3 2030	Legacy CED oscillators and clocks	338.0	71.1%	240.4
Q3 2030	Legacy auto industrial aerospace defense	71.5	64.0%	45.7
Q3 2030	Legacy mobile IoT consumer	88.5	60.0%	53.1
Q3 2030	Acquired Renesas TPD	243.1	55.0%	133.7
Q4 2030	Legacy CED oscillators and clocks	364.0	71.3%	259.4
Q4 2030	Legacy auto industrial aerospace defense	74.2	64.0%	47.5
Q4 2030	Legacy mobile IoT consumer	85.5	60.0%	51.3
Q4 2030	Acquired Renesas TPD	261.8	56.3%	147.4

Operating-cost derivation

Quarter	Cash-basis R&D incl. ordinary depreciation and rent	Cash-basis SG&A incl. ordinary depreciation and rent	Operating SBC; COGS SBC is already in product GP	Integration and transition provision
Q3 2026	42.9	39.6	35.0	6.0
Q4 2026	44.4	41.0	35.9	6.0
Q1 2027	46.0	42.4	36.8	4.0
Q2 2027	47.6	43.9	37.7	4.0
Q3 2027	49.2	45.4	38.6	4.0
Q4 2027	51.0	47.0	39.6	4.0
Q1 2028	52.7	48.7	40.6	2.0
Q2 2028	54.6	50.4	41.6	2.0
Q3 2028	56.5	52.1	42.6	2.0
Q4 2028	58.5	54.0	43.7	2.0
Q1 2029	60.5	55.9	44.8	2.0
Q2 2029	62.6	57.8	45.9	2.0
Q3 2029	64.8	59.8	47.1	2.0
Q4 2029	67.1	61.9	48.2	2.0
Q1 2030	69.4	64.1	49.5	2.0
Q2 2030	71.9	66.3	50.7	2.0
Q3 2030	74.4	68.7	52.0	2.0
Q4 2030	77.0	71.1	53.3	2.0

The categories reconcile to reference operating expenses. Incremental costs below are applied once; depreciation or stock compensation already included in a reference category is not added again.

Probability-weighted financial bridge

Active = reference + sum(probability x bounded conditional delta). Negative deltas are retained. A branch with zero near-term increment remains zero only for the timing or overlap reasons documented in the evidence register.

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q3 2026	SITM-TPD	70%	0.0	0.0	0.0	0.0
Q3 2026	SITM-SYNC	65%	0.0	0.0	0.0	0.0
Q3 2026	SITM-REPRICE	45%	0.0	0.0	0.0	0.0
Q3 2026	SITM-VTD	35%	0.0	0.0	0.0	0.0
Q3 2026	SITM-CYCLE	25%	0.0	0.0	0.0	0.0
Q4 2026	SITM-TPD	70%	5.0	3.6	0.9	1.9
Q4 2026	SITM-SYNC	65%	0.0	0.0	0.0	0.0
Q4 2026	SITM-REPRICE	45%	0.0	0.0	0.0	0.0
Q4 2026	SITM-VTD	35%	0.0	0.0	0.0	0.0
Q4 2026	SITM-CYCLE	25%	0.0	0.0	0.0	0.0
Q1 2027	SITM-TPD	70%	20.0	14.6	3.6	7.7
Q1 2027	SITM-SYNC	65%	6.9	5.1	1.5	2.3
Q1 2027	SITM-REPRICE	45%	0.8	0.7	0.0	0.3
Q1 2027	SITM-VTD	35%	0.0	0.0	0.0	0.0
Q1 2027	SITM-CYCLE	25%	-28.0	-19.9	-3.4	-4.1
Q2 2027	SITM-TPD	70%	28.0	20.4	5.0	10.8
Q2 2027	SITM-SYNC	65%	7.6	5.6	1.7	2.6
Q2 2027	SITM-REPRICE	45%	0.9	0.7	0.0	0.3
Q2 2027	SITM-VTD	35%	0.0	0.0	0.0	0.0
Q2 2027	SITM-CYCLE	25%	-30.5	-21.7	-3.7	-4.5
Q3 2027	SITM-TPD	70%	36.0	26.3	6.5	13.9
Q3 2027	SITM-SYNC	65%	8.2	6.1	1.8	2.8
Q3 2027	SITM-REPRICE	45%	1.0	0.8	0.0	0.3
Q3 2027	SITM-VTD	35%	0.0	0.0	0.0	0.0
Q3 2027	SITM-CYCLE	25%	-33.1	-23.5	-4.0	-4.9
Q4 2027	SITM-TPD	70%	42.0	30.7	7.6	16.2
Q4 2027	SITM-SYNC	65%	8.8	6.5	1.9	3.0
Q4 2027	SITM-REPRICE	45%	1.0	0.8	0.1	0.4

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q4 2027	SITM-VTD	35%	0.0	0.0	0.0	0.0
Q4 2027	SITM-CYCLE	25%	-35.6	-25.3	-4.3	-5.3
Q1 2028	SITM-TPD	70%	40.0	29.2	7.2	15.4
Q1 2028	SITM-SYNC	65%	13.9	10.3	3.0	4.7
Q1 2028	SITM-REPRICE	45%	1.1	0.9	0.1	0.4
Q1 2028	SITM-VTD	35%	4.0	2.7	3.0	-0.1
Q1 2028	SITM-CYCLE	25%	-36.8	-26.1	-4.4	-5.4
Q2 2028	SITM-TPD	70%	44.0	32.1	7.9	16.9
Q2 2028	SITM-SYNC	65%	15.1	11.2	3.3	5.1
Q2 2028	SITM-REPRICE	45%	1.2	0.9	0.1	0.4
Q2 2028	SITM-VTD	35%	4.3	2.9	3.1	-0.0
Q2 2028	SITM-CYCLE	25%	-40.2	-28.5	-4.8	-5.9
Q3 2028	SITM-TPD	70%	47.0	34.3	8.5	18.1
Q3 2028	SITM-SYNC	65%	16.4	12.1	3.6	5.5
Q3 2028	SITM-REPRICE	45%	1.3	1.0	0.1	0.4
Q3 2028	SITM-VTD	35%	4.7	3.2	3.2	0.0
Q3 2028	SITM-CYCLE	25%	-43.5	-30.9	-5.2	-6.4
Q4 2028	SITM-TPD	70%	49.0	35.8	8.8	18.9
Q4 2028	SITM-SYNC	65%	17.6	13.1	3.9	6.0
Q4 2028	SITM-REPRICE	45%	1.4	1.1	0.1	0.5
Q4 2028	SITM-VTD	35%	5.0	3.4	3.3	0.1
Q4 2028	SITM-CYCLE	25%	-46.9	-33.3	-5.6	-6.9
Q1 2029	SITM-TPD	70%	48.0	35.0	8.6	18.5
Q1 2029	SITM-SYNC	65%	20.0	14.8	4.4	6.8
Q1 2029	SITM-REPRICE	45%	1.4	1.1	0.1	0.5
Q1 2029	SITM-VTD	35%	11.9	8.1	5.0	1.1
Q1 2029	SITM-CYCLE	25%	-47.3	-33.6	-5.7	-7.0
Q2 2029	SITM-TPD	70%	51.0	37.2	9.2	19.6
Q2 2029	SITM-SYNC	65%	21.8	16.2	4.8	7.4
Q2 2029	SITM-REPRICE	45%	1.5	1.2	0.1	0.5
Q2 2029	SITM-VTD	35%	13.0	8.8	5.2	1.3

Quarter	Branch	p	Conditional revenue	Conditional GP	Conditional opex	Weighted OP
Q2 2029	SITM-CYCLE	25%	-51.6	-36.6	-6.2	-7.6
Q3 2029	SITM-TPD	70%	54.0	39.4	9.7	20.8
Q3 2029	SITM-SYNC	65%	23.7	17.5	5.2	8.0
Q3 2029	SITM-REPRICE	45%	1.6	1.3	0.1	0.5
Q3 2029	SITM-VTD	35%	14.0	9.5	5.5	1.4
Q3 2029	SITM-CYCLE	25%	-55.8	-39.7	-6.7	-8.2
Q4 2029	SITM-TPD	70%	57.0	41.6	10.3	21.9
Q4 2029	SITM-SYNC	65%	25.5	18.9	5.6	8.6
Q4 2029	SITM-REPRICE	45%	1.7	1.4	0.1	0.6
Q4 2029	SITM-VTD	35%	15.1	10.3	5.8	1.6
Q4 2029	SITM-CYCLE	25%	-60.1	-42.7	-7.2	-8.9
Q1 2030	SITM-TPD	70%	54.0	39.4	9.7	20.8
Q1 2030	SITM-SYNC	65%	26.4	19.5	5.8	8.9
Q1 2030	SITM-REPRICE	45%	1.7	1.4	0.1	0.6
Q1 2030	SITM-VTD	35%	23.8	16.2	7.9	2.9
Q1 2030	SITM-CYCLE	25%	-59.0	-41.9	-7.1	-8.7
Q2 2030	SITM-TPD	70%	58.0	42.3	10.4	22.3
Q2 2030	SITM-SYNC	65%	28.8	21.3	6.3	9.7
Q2 2030	SITM-REPRICE	45%	1.9	1.5	0.1	0.6
Q2 2030	SITM-VTD	35%	25.9	17.6	8.5	3.2
Q2 2030	SITM-CYCLE	25%	-64.4	-45.7	-7.7	-9.5
Q3 2030	SITM-TPD	70%	62.0	45.3	11.2	23.9
Q3 2030	SITM-SYNC	65%	31.2	23.1	6.9	10.5
Q3 2030	SITM-REPRICE	45%	2.0	1.6	0.1	0.7
Q3 2030	SITM-VTD	35%	28.1	19.1	9.0	3.5
Q3 2030	SITM-CYCLE	25%	-69.7	-49.5	-8.4	-10.3
Q4 2030	SITM-TPD	70%	66.0	48.2	11.9	25.4
Q4 2030	SITM-SYNC	65%	33.6	24.9	7.4	11.4
Q4 2030	SITM-REPRICE	45%	2.2	1.7	0.1	0.7
Q4 2030	SITM-VTD	35%	30.2	20.6	9.6	3.9
Q4 2030	SITM-CYCLE	25%	-75.1	-53.3	-9.0	-11.1

Coherent operating and funding outcomes

Five event outcomes form 32 mutually exclusive combinations. Each outcome persists across the quarterly path. Event independence is a model assumption. Common equity equals the positive part of 20 times state-specific four-quarter operating profit less financial claims outstanding at the valuation quarter, divided by shares at the end of the earnings window. The floor is applied before probability weighting; cash is not added or used to offset this debt.

Event-bit order: SITM-TPD, SITM-SYNC, SITM-REPRICE, SITM-VTD, SITM-CYCLE. 1 means that event occurs; 0 means it does not. A 1 in the final position is the adverse execution event.

State / bits	Probability	Q4 2027 F4Q OP	Q4 2027 ceiling	Minimum path cash
S00 / 00000	2.82%	405.2	207.13	424.2
S01 / 00001	0.94%	313.1	150.28	424.2
S02 / 00010	1.52%	404.8	206.86	424.2
S03 / 00011	0.51%	312.6	150.01	424.2
S04 / 00100	2.30%	408.6	209.23	424.2
S05 / 00101	0.77%	316.5	152.38	424.2
S06 / 00110	1.24%	408.2	208.96	424.2
S07 / 00111	0.41%	316.0	152.11	424.2
S08 / 01000	5.23%	433.4	224.51	424.2
S09 / 01001	1.74%	341.2	167.66	424.2
S10 / 01010	2.82%	432.9	224.25	424.2
S11 / 01011	0.94%	340.8	167.40	424.2
S12 / 01100	4.28%	436.8	226.61	424.2
S13 / 01101	1.43%	344.6	169.76	424.2
S14 / 01110	2.30%	436.3	226.35	424.2
S15 / 01111	0.77%	344.2	169.50	424.2
S16 / 10000	6.57%	500.3	265.84	424.2
S17 / 10001	2.19%	408.2	208.99	424.2
S18 / 10010	3.54%	499.9	265.58	424.2
S19 / 10011	1.18%	407.8	208.73	424.2
S20 / 10100	5.37%	503.7	267.94	424.2
S21 / 10101	1.79%	411.6	211.09	424.2
S22 / 10110	2.89%	503.3	267.68	424.2
S23 / 10111	0.96%	411.2	210.83	424.2
S24 / 11000	12.20%	528.5	283.23	424.2
S25 / 11001	4.07%	436.4	226.38	424.2

State / bits	Probability	Q4 2027 F4Q OP	Q4 2027 ceiling	Minimum path cash
S26 / 11010	6.57%	528.1	282.97	424.2
S27 / 11011	2.19%	436.0	226.12	424.2
S28 / 11100	9.98%	531.9	285.33	424.2
S29 / 11101	3.33%	439.8	228.48	424.2
S30 / 11110	5.37%	531.5	285.07	424.2
S31 / 11111	1.79%	439.4	228.22	424.2

Minimum path cash is the lowest quarter-end cash balance from Q3 2026 through Q4 2030 in that coherent state. It is a funding check, not an extra valuation asset. All 32 modeled paths remain above the minimum liquidity policy.

Quarterly debt, shares and funding envelope

Conditional financing outputs are retained with each operating state. The compact table shows the full path without repeating 576 state-quarter records. Shares below are each operating quarter; each ceiling uses the share count three quarters later.

Quarter	Financial claims	Quarter shares	Expected cash	Lowest state cash	Highest state capex
Q3 2026	1,431.4	31.450	424.2	424.2	17.4
Q4 2026	1,423.2	31.570	478.3	477.7	19.6
Q1 2027	1,415.1	31.690	536.6	527.0	20.2
Q2 2027	1,406.9	31.810	601.0	573.6	22.5
Q3 2027	1,398.8	31.930	680.3	631.9	25.5
Q4 2027	1,390.7	32.050	773.6	701.4	27.2
Q1 2028	1,382.5	32.170	870.6	771.8	27.4
Q2 2028	1,374.4	32.290	976.3	849.8	30.0
Q3 2028	1,366.3	32.410	1,102.2	945.6	33.4
Q4 2028	1,358.1	32.530	1,244.9	1,056.4	35.2
Q1 2029	1,350.0	32.650	1,395.1	1,173.8	34.8
Q2 2029	1,350.0	32.770	1,554.2	1,299.4	38.0
Q3 2029	1,350.0	32.890	1,738.2	1,445.4	42.1
Q4 2029	1,350.0	33.010	1,944.0	1,610.4	44.5
Q1 2030	1,350.0	33.130	2,145.1	1,769.5	43.0
Q2 2030	1,350.0	33.250	2,354.1	1,934.6	46.9
Q3 2030	1,350.0	33.370	2,593.2	2,126.1	51.9
Q4 2030	1,350.0	33.490	2,858.7	2,340.5	55.1

Claim decisions, overlap and triggers

SITM-TPD

claim: September TPD growth 60-80% for 2027, constrained by Renesas supply/TSA

source: September 9 issuer conference p10

probability: 0.7

probability rationale: Specific management range and an already shipped business support above-average probability; supply limits and integration prevent certainty.

Conditional outcome: 2027 TPD revenue $566m vs $440m reference; conditional increment $126m. Bound $80-160m. Q3 2026 remains the $85m guide.

overlap: Reference already includes $440m, approximately32% growth over the modeled standalone 2026 level. The $300m old forecast and $85m quarterly run-rate are not additional vectors.

cost bridge: 73% incremental GM less18% additional development/sales/support =55% incremental OP.

active contribution: 2027 +$88.2m revenue and +$48.51m OP; quarterly ramp20/28/36/42 before70% weighting.

duration: 2027-30; incremental dollars fade relative to total growth

trigger: Raise to80% on two quarters of shipments meeting ramp without receivable/inventory stress. Reduce to40% on two-quarter TSA or test capacity delay.

SITM-SYNC

claim: Inference synchronization and more timing content per qualified rack

source: Q1/Q2 calls and September9 conference

probability: 0.65

probability rationale: Products exist and design wins span hyperscalers, but rack count, mix and realized price are not disclosed.

Conditional outcome: 2027 150,000 qualified rack-equivalents x$300 additional content x70% non-overlap=$31.5m. Bounds100,000-200,000 and$200-400, with capture constrained by existing CED revenue.

overlap: 30% overlap removed; transceivers, CPO and the cumulativeElite2SAM are not stacked as independent TAMs. Rack-equivalents are an analytical normalization, not disclosed physical shipments.

cost bridge: 74% incremental GP less22% applications engineering/sales expenses yields52% OP.

active contribution: 2027 +$20.475m revenue and +$10.647m OP.

trigger: Validate Elite2 production and realized CED mix; increase to75% on volume evidence, cut to40% on slipped qualifications or customer value unproven.

SITM-REPRICE

claim: Premiums for qualified short-lead-time timing supplies

source: Q1 supply/performance premium and Q2/September TPD constraints

probability: 0.45

probability rationale: A documented premium franchise and customer delay costs support some repricing; no disclosed general price rise.

Conditional outcome: 2% premium on60% eligible legacy CED, less50% already in richer mix. Bound0-3%.

overlap: TPD excluded because its growth/margin branch already addresses the constrained supply. No separate utilization uplift is added.

cost bridge: 80% incremental GP after expediting/yield/service costs;5% additional selling support.

active contribution: 2027 +$1.674m revenue and +$1.2555m OP.

trigger: Raise to60% with realized ASP evidence excluding mix; reduce to20% if quartz competition or volume discounts offset premium.

SITM-VTD

claim: Embedded timing/chiplet/substrate systems from2028-29

source: September9 CEO $1-1.5bn incremental SAM discussion, p6

probability: 0.35

probability rationale: Physical signal-integrity rationale and disclosed roadmap justify positive probability before named volume programs; packaging qualification remains uncertain.

Conditional outcome: 2028 $18m net incremental revenue =60m eligible new-socket wallet x50%capture x60% non-overlap; $54m2029/$108m2030. Bound2028 $0-30m.

overlap: 40% removed for replacement of discrete timing and reference CED growth. September narrower $1-1.5bn statement governs; no sum with earlier $2.5bn framing.

cost bridge: 68% GP less25% variable engineering/packaging support and$8m annual launch expense.

active contribution: 2028 +$6.3m revenue and -$0.091m OP; early investment is retained even where launch profit is negligible.

trigger: Raise to50% on working package qualifications and named roadmap timing; reduce to15% if launch moves past2029.

SITM-CYCLE

claim: AI customer digestion, concentration and capacity response

source: Q2 risks; Q1 CEO acknowledges credible quartz suppliers

probability: 0.25

probability rationale: Strong forward demand reduces but does not remove semiconductor/customer-cycle risk.

Conditional outcome: 12% shortfall in reference legacyCED plusTPD from2027; bound8-20%.

overlap: Applies only to reference businesses; no second cut to separately weighted new-socket revenues.

cost bridge: 71% decremental GP with only12% expense relief; engineering/SBC/PPA remain.

active contribution: 2027 -$31.8m revenue and -$18.762m OP.

trigger: Reduce to15% with broad repeat orders and stable inventory; increase to45% if two large programs delay or price competition accelerates.

SITM-TECH

claim: Article says quartz is finished and MEMS only recently crossed below100fs

probability: 0.1

Conditional outcome: Sector-wide exclusion of quartz by2027 is implausible under the current evidence.

overlap: Product performance and qualification advantage already feed theSYNCbranch and existing reference margins.

cost bridge: Zero additional monopoly margin or TAM award. The CEO explicitly describes credible1.6T competitors;70fs was public in2020 and4-16A revision predates2026.

active contribution: Zero extra OP; no incremental quantifiable benefit beyond the already modeled differentiated-product ramp.

trigger: Reassess only with matched part tests, production supplier removals and sustainable realized economics.

SITM-OTHER

claim: Defense doubling, modem adoption, LEO and humanoid opportunities

probability: 0.6

Conditional outcome: ReferenceAID$150m in2027 to $275m2030;MIC$220m to $295m. Modem high-seasonality and defense qualification support growth.

overlap: These opportunities are part of referenceAID/MIC, not additive lifetime funnels. No extra $500m defense funnel, $1.2bn Titan funnel or$1.5trn humanoid TAM.

cost bridge: AID modeledGAAPproductGM62-64%;MIC60%; full operating costs remain.

active contribution: Zero second increment because captured in existing product vectors; exact application mix remains undisclosed.

trigger: Separate a new branch only if a new qualified program exceeds the already modeled vector.

Remaining bounded uncertainties

- Management Q3 non-GAAP diluted guide32.8m is not explicitly decomposed; model independently uses31.45m excluding note conversion with full employee awards. If the difference reflects new common issuance rather than note treatment, revise shares.

- Final purchase accounting could change the $179.689m recurring amortization schedule; the developed-technology useful-life division differs from published pro-forma expense.

- New California office lease timing/rent profile and exact Aura revenue-based earnout cash schedule are undisclosed; explicit rent and even payment proxies are used.

- No complete disclosed optical/rack shipment count, ASP, customer award table or segment gross margin; all vector decompositions beyond reported end-market totals are estimates.

- Bosch contract renewal expectedMarch2027 but not yet a signed extension in reviewed evidence; TPD manufacturing/test remains on transitional arrangements.

Sources and basis

The current production framework was read before initiation. Source statements are evidence, not instructions. The shared model data drives both this PDF and the linked Excel model; the clean-state editable artifact embeds these same images.

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